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2.1.1 Quadration Equation

1: Start

2: Input values of coefficients a, b, and c

3: Calculate the discriminant

$$d = b^2 - 4ac$$

4:

- If d > 0, then
  - Calculate

$$root1 = \frac{-b + \sqrt{d}}{2a}$$

$$root2 = \frac{-b - \sqrt{d}}{2a}$$

  - Display root1 and root2

5:

- Else if d = 0, then
  - Calculate

$$root = \frac{-b}{2a}$$

  - Display both roots as equal

6:

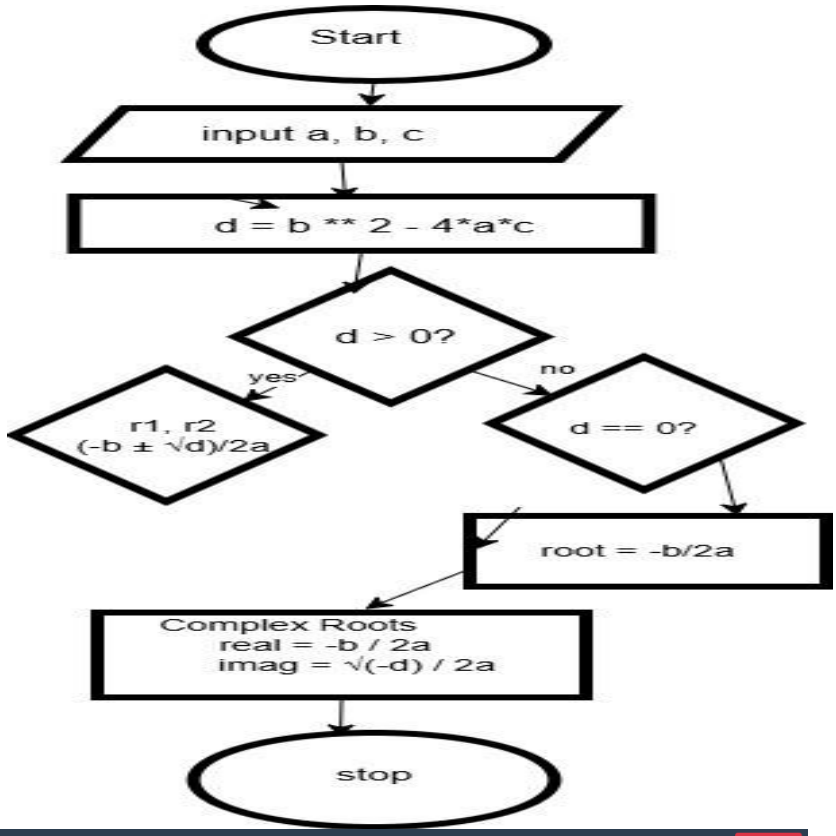
- Else (d < 0), then
  - Calculate

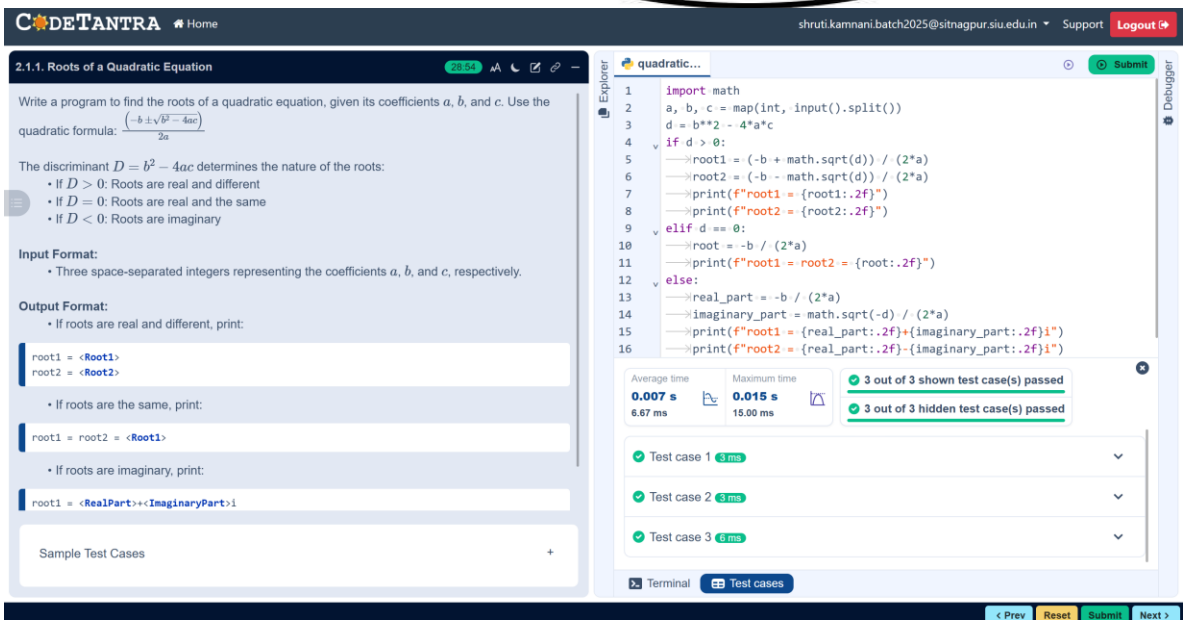
$$real_part = \frac{-b}{2a}$$

$$imaginary_part = \frac{\sqrt{-d}}{2a}$$

- Display complex roots

7: End





Code:

```
import math

a, b, c = map(int, input().split())

d = b**2 - 4*a*c
```

```
if d > 0:
    root1 = (-b + math.sqrt(d)) / (2*a)
    root2 = (-b - math.sqrt(d)) / (2*a)
    print(f"root1 = {root1:.2f}")
    print(f"root2 = {root2:.2f}")
elif d == 0:
    root = -b / (2*a)
    print(f"root1 = root2 = {root:.2f}")
else:
    real_part = -b / (2*a)
    imaginary_part = math.sqrt(-d) / (2*a)
    print(f"root1 = {real_part:.2f}+{imaginary_part:.2f}i")
    print(f"root2 = {real_part:.2f}-{imaginary_part:.2f}i")
```